

Evolutionary Computation and its Applications

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Today's goals

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Module 1

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Fundamental Concepts in Optimization

Module 2

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Genetic Algorithm

Module 5 Introduction

✔ Introduction to Module 5's Study

Video • 1 min

✔ How to Access MATLAB Online

Reading • 5 min

Module 5-1 Genetic Algorithm 1

✔ Genetic Algorithm: A First Introduction

Video • 6 min

✔ Genetic Algorithm: A Demo

Video • 8 min

✔ Materials on Genetic Algorithms

Reading • 10 min

✔ How Genetic Algorithms Solve Optimization Problems

Reading • 5 min

Module 5-2 Genetic Algorithm 2

Genetic Algorithms: GA Representation, Initialization and Selection

Video • 10 min

Genetic Algorithms: GA Crossover and Mutation

Video • 7 min

Genetic Algorithms: GA Stopping Criterion and Summary

Video • 5 min

Materials on Genetic Algorithms

Reading • 10 min

How GA Operators Work Together

Reading • 5 min

Module 5-3 Use Genetic Algorithm 1

Use GA via MATLAB: GA Function

Video • 7 min

Use GA via MATLAB: Employ the Function

Video • 6 min

Materials on Using Genetic Algorithms

Reading • 10 min

Using Genetic Algorithms Effectively in MATLAB

Reading • 5 min

Module 5-4 Use Genetic Algorithm 2

Use GA via MATLAB: Case Studies

Video • 5 min

Materials on Case Studies

Reading • 10 min

Solving Real Engineering Problems with Genetic Algorithms

Reading • 5 min

Module 5 Quiz

Module 5 Quiz

Practice Assignment • 30 min

Module 5 Assignment

Module 5 Assignment

Graded App Item • 30 min

Objective function for Assignment 5

Reading • 5 min

Applying Genetic Algorithms to a Real Engineering Design Problem

Reading • 5 min

Designing and Applying a Genetic Algorithm to Solve an Engineering Optimization Problem

Dialogue • 5 min

Module 5 Recap

Reading • 5 min

Module 3

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Particle Swarm Optimization

How Genetic Algorithms Solve Optimization Problems

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In this section, you explored genetic algorithms and how principles inspired by natural evolution can be applied to optimization problems.

Now, reflect on your understanding:

- Why do you view a genetic algorithm as an AI technique rather than just a conventional optimization method? What characteristics make it adaptive or intelligent in the way it searches for solutions?
- How do the key components—population, selection, crossover, and mutation—interact in your understanding to progressively improve solution quality over iterations?
- In the demonstration example, why does the average fitness of the population tend to increase across generations, even when some individual solutions perform worse? What does this suggest about the overall dynamics of the search process?
- Based on what you have learned so far, what advantages do genetic algorithms offer when dealing with complex, non-linear, or multimodal optimization problems?

This reflection encourages you to build a clearer mental model of how genetic algorithms function before diving deeper into the details of individual operators.

Go to next item

✔ Completed

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